

PROJECT DOCUMENTATION

Mineral Quality Assessment & Economic Feasibility Decision-Support System

Placement & Technical Interview Reference

Technology: Python • Streamlit • Pandas • NumPy • SQLite • ReportLab

1. Project Overview

This project is a Python-based decision-support application for assessing mineral/ore quality together with economic feasibility. The application accepts chemical, physical and economic sample information, converts selected quality indicators into normalized scores, combines them using configurable weights, checks specification compliance, classifies the sample, and provides statistical, validation and reporting tools.

The system is intended as a decision-support prototype. Its project-defined quality classes and scoring model are not universal industry standards and should be used with verified specifications, traceable laboratory/field data and professional judgement.

2. Problem Statement

Mineral quality assessment can involve several parameters such as grade, moisture, pH, chemical behaviour and homogeneity. Economic feasibility also depends on selling price and multiple cost components. Evaluating these factors manually can make comparison, record keeping and reproducibility difficult. The project provides a structured software workflow that combines these inputs into an auditable assessment.

3. Objectives

- Develop a user-friendly Streamlit application for mineral sample assessment.
- Store sample records and calculation history using SQLite.
- Normalize multiple quality indicators to a common 0–100 scale.
- Calculate a configurable weighted overall score and project-defined quality class.
- Check grade, moisture and pH against configured specifications.
- Estimate economic feasibility using selling price and cost components.
- Support Excel/CSV data import and export of results.
- Provide descriptive statistics, correlations and sensitivity analysis.
- Record data sources, test methods, specification references and validation information.
- Generate PDF reports for individual sample assessments.

4. Technology Stack

Technology	Purpose	Why Used
Python	Core programming	Rapid development and scientific/data-processing ecosystem
Streamlit	Web application UI	Builds an interactive app directly in Python
Pandas	Data processing	Tabular data, statistics, Excel/CSV

		workflows
NumPy	Numerical operations	Efficient numerical calculations
SQLite	Database	Lightweight local persistent storage
ReportLab	PDF generation	Creates downloadable assessment reports
Excel/CSV	Data exchange	Bulk import and template-based data entry

5. Application Modules

Module	Function
New Sample	Enter a mineral sample, quality parameters, economic inputs and evidence fields.
Dashboard	View sample count, average score, quality-class distribution and mineral-wise summaries.
Statistics	Descriptive statistics, coefficient of variation, correlations and scatter analysis.
Excel Import	Import CSV/Excel datasets, validate rows and store valid records.
Economics	Calculate margins and examine economic sensitivity to price/cost changes.
Specifications	Maintain mineral specifications and versioned scoring weights.
Validation & Evidence	Compare application classes with independently entered reference classes and show a confusion matrix.
Data Quality	Audit missing values, evidence coverage and data quality issues.
Methodology	Explain scoring assumptions, quality classes and reproducibility.
Project Information	Describe scope, technology and limitations.

6. Assessment Methodology

Seven assessment dimensions are converted to normalized scores from 0 to 100:

1. Grade
2. pH
3. Chemical Stability
4. Chemical Reactivity
5. Moisture
6. Homogeneity
7. Economic Feasibility

7. Weighted Scoring Model

The default project weights are:

Dimension	Weight
Grade	25%
pH	10%

Chemical Stability	10%
Chemical Reactivity	10%
Moisture	10%
Homogeneity	10%
Economic Feasibility	25%

$$\text{Overall Score} = \sum(\text{score}_i \times \text{weight}_i) / \sum(\text{weight}_i)$$

These weights are configurable project choices. In an academic presentation, explain that the weights must be justified for the selected mineral/application rather than presented as universal industry values.

8. Quality Classification

- HIGH: 85–100
- GOOD: 70–84.99
- MODERATE: 50–69.99
- LOW: below 50

Important: these boundaries are project-defined classification thresholds. They should not be described as universal industrial quality grades.

9. Economic Feasibility

The economic component considers selling price and total cost. Total cost is represented by mining cost, processing cost, transport cost and other cost components.

$$\text{Total Cost} = \text{Mining Cost} + \text{Processing Cost} + \text{Transport Cost} + \text{Other Cost}$$

$$\text{Margin} = \text{Selling Price} - \text{Total Cost}$$

The application converts the economic result into a normalized economic score. The project methodology uses a break-even reference corresponding to an economic score of 50.

10. Data and Database Design

SQLite is used for persistent local storage. A sample record stores identification and traceability information, measured parameters, individual component scores, overall score, quality class, compliance status, economic values, specification information and validation evidence.

- Sample ID, date, mineral, location and batch/lot
- Main component and grade; impurity information
- pH, moisture, stability, reactivity and homogeneity
- Selling price and cost components
- Individual normalized scores and overall score
- Quality class and specification compliance
- Specification source/reference and specification version
- Scoring-weight version
- Data source, test method and validation/reference class

11. Reproducibility and Versioning

A major project feature is reproducibility. Each saved sample records the specification version and scoring-weight version used at the time of calculation. Therefore, changing a later specification or weight does not silently rewrite the historical basis of an older result.

12. Validation Strategy

The application can accept an independently obtained reference quality class for a sample. It compares the application class with the reference class, reports classification agreement and displays a confusion matrix.

This is comparative validation evidence, not proof of scientific or laboratory validity. For the final academic project, validation should use traceable real data and documented reference criteria.

13. Data Quality Controls

- Missing-value audit
- Range and input validation
- Duplicate/sample record checks where applicable
- Evidence coverage for data source, test method and specification reference
- Rejected-row reporting during Excel/CSV import
- Explicit labelling of synthetic demo data

14. Typical Workflow

8. Collect a mineral sample and record its source and test method.
9. Enter the sample manually or prepare an Excel/CSV dataset.
10. Select the appropriate mineral specification.
11. Enter grade, pH, moisture, chemical behaviour, homogeneity and economic inputs.
12. Normalize the individual indicators.
13. Apply the configured weights to calculate the overall score.
14. Assign the project-defined quality class.
15. Check specification compliance.
16. Review economic feasibility and sensitivity.
17. Compare against independent reference data when available.
18. Review data-quality/audit results and export the report.

15. Key Challenges Solved

Multi-parameter assessment: Combined different indicators on a common normalized scale.

Subjectivity in weighting: Made weights configurable and versioned instead of hard-coding one permanent model.

Traceability: Stored sample source, test method and specification references.

Historical reproducibility: Stored specification and weight versions with each result.

Bulk data entry: Added Excel/CSV import with row validation.

Decision transparency: Displayed individual scores and methodology rather than only a final label.

Validation: Added reference-class comparison and confusion-matrix reporting.

Reporting: Added downloadable PDF reports and dataset export.

16. What I Would Say in an Interview

30-second answer

“My project is a Python and Streamlit based mineral quality and economic feasibility decision-support system. It accepts mineral quality and cost data, normalizes multiple parameters to 0–100 scores, applies configurable weights, checks specification compliance, and produces an overall project-defined quality classification. I also implemented SQLite storage, Excel import, statistical analysis, economic sensitivity, validation against independent reference classes and PDF reporting.”

60–90 second answer

“The main problem I addressed was that mineral assessment involves several quality and economic parameters, so I wanted a structured and reproducible workflow instead of a manual spreadsheet-only process. I developed the application in Python using Streamlit for the interface, Pandas and NumPy for data processing, SQLite for persistent storage and ReportLab for PDF reports. Seven dimensions are normalized to a 0–100 scale and combined through configurable weights. The application also checks grade, moisture and pH against configured specifications, calculates economic margin, supports bulk Excel import, performs statistical analysis and sensitivity analysis, and stores specification and weight versions for reproducibility. For validation, the system can compare its classification with an independently obtained reference class. The important limitation is that the score and quality classes are project-defined decision-support outputs and do not replace certified laboratory testing or applicable standards.”

17. Common Technical Interview Questions and Answers

Q: Why did you choose Python?

A: Python provides strong libraries for data processing, numerical computation, database access and rapid application development. It allowed the complete prototype to be built in one ecosystem.

Q: Why Streamlit instead of a traditional web framework?

A: The project is data- and analytics-oriented. Streamlit allows an interactive interface to be developed directly around Python data-processing code with relatively little front-end code.

Q: Why SQLite?

A: SQLite is lightweight, serverless and sufficient for a local academic prototype. It provides persistent structured storage without requiring a separate database server.

Q: Why normalize the parameters?

A: The input parameters have different units and ranges. Normalization converts them to a common scale so that a weighted combination is meaningful.

Q: How is the overall score calculated?

A: Each normalized score is multiplied by its configured weight, the weighted values are summed, and the result is divided by the total weight.

Q: Why are the weights configurable?

A: Different minerals or applications may prioritize different factors. Configurability makes the model adaptable and allows sensitivity analysis.

Q: What is the purpose of specification versioning?

A: If a specification changes later, old results should still be reproducible using the criteria that were applied when the sample was assessed.

Q: How did you validate the application?

A: The application supports independent reference quality classes and calculates classification agreement plus a confusion matrix. The quality of validation depends on the independence and quality of the reference data.

Q: What is sensitivity analysis?

A: It examines how economic or scoring outputs change when important inputs such as selling price or cost assumptions are varied.

Q: What happens if an Excel row is invalid?

A: The import workflow attempts to build each record separately, accepts valid rows and reports rejected rows with the row number and error reason.

Q: What are the limitations?

A: The project-defined weights and class thresholds require justification, real applications require verified specifications and traceable data, and the system is a decision-support prototype rather than a replacement for certified laboratory or regulatory assessment.

Q: What did you personally learn?

A: I strengthened Python programming, data cleaning, statistical analysis, database handling, UI development, validation thinking, reproducibility and technical documentation.

18. Resume / LinkedIn Project Description

Mineral Quality Assessment & Economic Feasibility Decision-Support System | Python, Streamlit, Pandas, NumPy, SQLite, ReportLab

- Developed an interactive Streamlit application to assess mineral quality using normalized multi-parameter scoring and configurable weighted criteria.
- Implemented SQLite-based persistent storage with specification and scoring-weight versioning for reproducible historical assessments.
- Built Excel/CSV import, data-quality auditing, statistical analysis, economic sensitivity analysis, validation comparison and PDF reporting modules.

- Integrated specification compliance checks for key quality parameters and traceability fields for source, test method and reference information.

19. Questions You Should Be Ready For

- Explain the complete data flow from sample entry to final classification.
- Derive the weighted-average formula used in your application.
- Explain why each parameter is included.
- Explain and defend the selected weights.
- Explain how you selected the class boundaries.
- Explain how you would validate the model using real laboratory data.
- Explain the difference between validation agreement and scientific validity.
- Explain your database schema and why SQLite was selected.
- Explain what happens when the specification changes.
- Explain how you would deploy the application for multiple users.
- Explain how you would improve the model with a larger real dataset.
- Explain what you would change if a domain expert disagreed with the current scoring assumptions.

20. Future Improvements

- Replace demo specifications with verified standards/customer specifications relevant to the selected mineral.
- Use a larger, traceable laboratory/field dataset.
- Obtain independent expert/reference classifications for stronger validation.
- Perform formal sensitivity/robustness studies on weights and thresholds.
- Add authentication and role-based access for multi-user deployment.
- Move from local SQLite to PostgreSQL or another server database for production use.
- Add automated unit tests and continuous validation of scoring functions.
- Deploy the application on a controlled server or cloud environment.
- Add model calibration or machine-learning comparison only if sufficient labelled data is available.

21. Important Academic Statement

For viva/interview discussions, describe the output as a “project-defined decision-support classification” rather than claiming that the system determines universal mineral quality. Be prepared to explain the origin of the specifications, weights, thresholds and validation dataset. Synthetic demo data should never be presented as laboratory measurements.

22. One-line Project Summary

A Python-based decision-support system that combines mineral quality indicators, specification compliance and economic feasibility into a reproducible, auditable assessment workflow.